Method 2: the helium atom energy is

This can be rewritten as

The first term on the right vanishes at minimum energy. The ground energy and radius

are respectively as in .

The particle crosses the lead plate upward. Its charge is positive.

Velocity is normal to the field; Lorentz force is in the paper. Magnetism changes momentum direction, not magnitude. Neglecting gas drag, in the steady field the path is a circular arc of constant radius. The equation of motion is

Here is charge, speed and momentum; is the angle turned in time ,

and is the curvature radius. Hence

; let , be entering and exiting momenta. In ,

substitute the data to obtain

For , use special relativity. From

we obtain

Let , be entering and exiting speeds. Insert and

into to obtain

The mean speed through lead is is the transit